Practice workbook · What makes a CPU faster: clock speed, cores and cache

Clock speed, number of cores, cache size/type and their effect on how fast instructions are processed, including benefits and drawbacks of a higher clock speed and why cache helps.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-06

1. Question 5 · part b.iii · 2 marks

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Worked-example retry · Attempt part b.iii only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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1. Question 5 · part b.iii · 2 marks

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Worked-example retry · Attempt part b.iii only.

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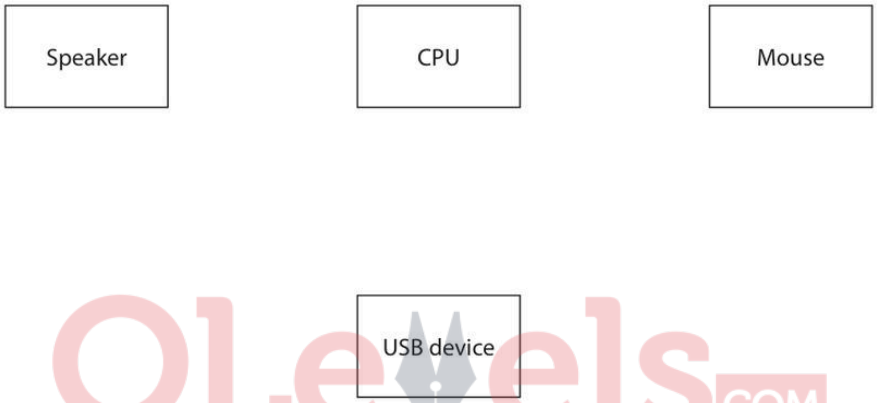
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(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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1. Question 5 · part b.iii · 2 marks

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Worked-example retry · Attempt part b.iii only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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1. Question 5 · part b.iii · 2 marks

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Worked-example retry · Attempt part b.iii only.

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(c) A washing machine uses several different embedded systems.

One embedded system uses a switch to identify the type of wash cycle selected by the user.

Describe **one other** example of an embedded system found in a washing machine.

(2)

(d) Programs use logic statements to control physical hardware.

(i) A window shuts when the temperature is too cool or it is a rainy night.

The values are defined as:

- A shows it is night time
- B shows it is too cool
- C shows it is raining.

Complete the truth table to show the results of each operation.

Two rows have been done for you.

(3)

A	B	C	A AND C	(A AND C) OR B
0	0	0	0	0
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1	1	1

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P 7 2 9 3 7 A 0 1 7 2 4

17

Turn over ▶

1. Question 5 · part b.iii · 2 marks · Working and answer

Working and answer

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2. Question 3 · part b · 2 marks

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Further practice · Attempt part b only.

3 The central processing unit (CPU) and other hardware components work together with the operating system to run programs and store data.

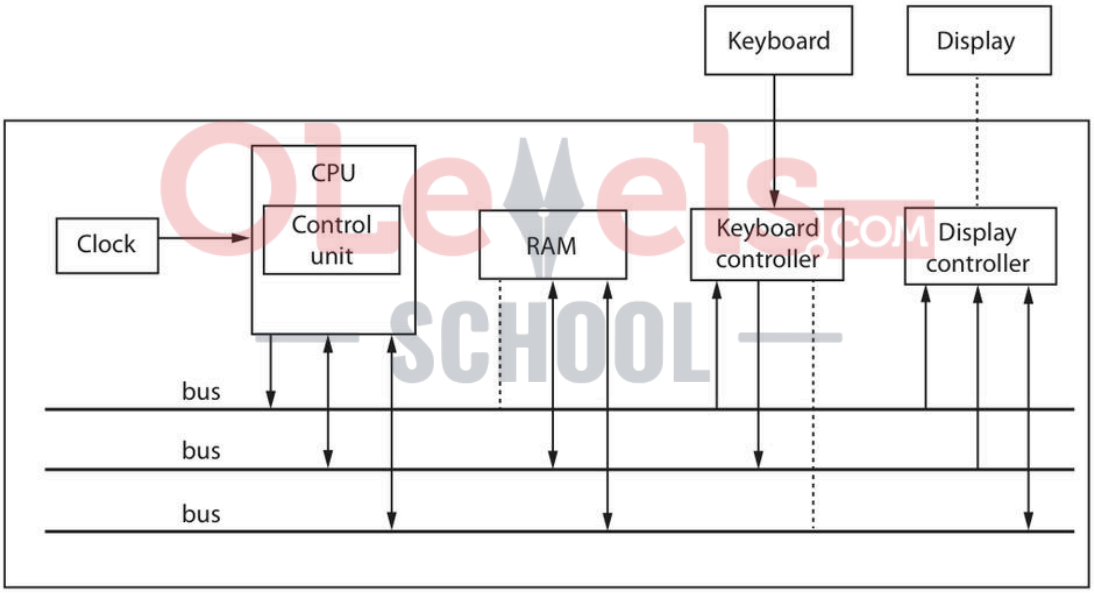
(a) Here is a partially completed diagram showing the hardware components used in the fetch-decode-execute cycle.

The function of the I/O controller is to manage communication between the CPU and input/output devices.

Complete the diagram by:

- labelling the buses
- adding directional arrows to the dotted lines to show the flow of communication.

(6)



(b) The CPU includes an area of cache memory.

Increasing the size of the cache will improve the performance of the CPU.

Explain how increasing the size of the cache improves the performance.

(2)

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P 6 1 8 8 4 R R A 0 8 2 0

2. Question 3 · part b · 2 marks

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Further practice · Attempt part b only.

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(c) Some computers have more than one CPU. The CPUs work in parallel to execute instructions.

Complete the table to show how two CPUs working in parallel could execute the expression:

(5 * 2) + (4 * 3) – (24 / 4)

(4)

CPU 1	CPU 2
5 * 2 = 10	

(d) One function of an operating system is to manage files on secondary storage.

Describe how an operating system manages the storage of a file on random access secondary storage.

(4)

(Total for Question 3 = 16 marks)

2. Question 3 · part b · 2 marks · Working and answer

Working and answer

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3. Question 3 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

3 Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

(2)

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

(2)

1 .....

2 .....

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

(2)

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

(2)

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3. Question 3 · part c · 2 marks

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Further practice · Attempt part c only.

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(d) Zakir uses many devices.

Complete the table by adding **one** tick (✓) in each row to show whether the device is an input device, an output device or both.

(2)

Device	Input	Output	Both
Headset			
Mouse			
Printer			
Touch screen			

(e) Zakir needs to protect his laptop against malware.

Give **two** types of anti-malware software he could install.

(2)

1 .....

2 .....

(Total for Question 3 = 12 marks)

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Working and answer

4. Question 5 · part b.iii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.iii only.

5 Computers can be general-purpose or specific-purpose machines.

(a) The central processing unit (CPU) is responsible for executing a program.

Figure 4 shows some of the hardware components of a computer system.

Complete the table by matching the component to the letters shown in **Figure 4**.

(5)

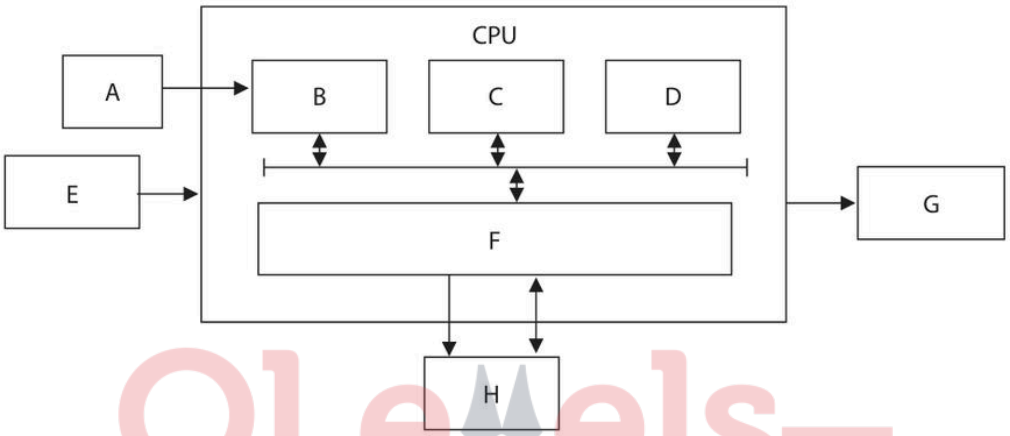


Figure 4

Component	Letter
Control unit	
Output device	
Registers	C
Main memory	
Cache	F
Input device	
Arithmetic/logic unit (ALU)	
Clock	A

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4. Question 5 · part b.iii · 1 mark

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Further practice · Attempt part b.iii only.

(b) The CPU runs a fetch-decode-execute cycle.

(i) Some registers within the CPU are general-purpose whilst others perform a specific function.

One register is the program counter. Its function is to keep track of the next instruction to be fetched.

Describe how it does this.

(2)

(ii) The control bus carries signals from the processor to other components.

Identify the signal that would be used during the **fetch** stage of the cycle.

(1)

- ☒ **A** Memory delete signal
- ☒ **B** Memory read signal
- ☒ **C** Memory store signal
- ☒ **D** Memory write signal

(iii) Clock speed is the number of pulses the CPU's clock generates per second.

State how increasing the speed of the clock impacts on the fetch-decode-execute cycle.

(1)

(c) An embedded system forms part of a larger system, device or machine.

(i) Give **two** features of an embedded system.

(2)

1

2

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P 6 9 2 9 7 A 0 1 6 2 0

4. Question 5 · part b.iii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part c.i · 1 mark

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Further practice · Attempt part c.i only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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5. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

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6. Question 1 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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6. Question 1 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

(d) Identify which **one** of these describes a sequential computational model.

(1)

- ☒ **A** Program instructions are read one after another from external storage
- ☒ **B** Program instructions are executed by multiple agents working together
- ☒ **C** Program instructions are executed in parallel by different cores
- ☒ **D** Program instructions are executed one after another

(e) A program can be written in a high-level or a low-level language.

(i) Give **one** reason for writing a program in a low-level language.

(1)

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.....

(ii) State the purpose of an assembler.

(1)

.....

(iii) Complete the table by adding **one** tick (✓) in **each** row to match the description.

(3)

Description	Compiler	Interpreter
Translates the program each time it is executed		
Produces permanent object code		
Translates line by line		
Translates the whole program before it is run		
Generates a list of errors once the complete program has been translated		

(Total for Question 1 = 11 marks)

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6. Question 1 · part c.ii · 1 mark · Working and answer

Working and answer

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7. Question 5 · part b · 2 marks

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Further practice · Attempt part b only.

5 A team is developing a computer game.

(a) Complete the table by adding a tick (✓) to show the characteristics of program language translators.

You may select more than one translator per characteristic.

(3)

Characteristic	Translator		
	Assembler	**Interpreter**	**Compiler**
Does **not** have to be present in memory to execute the translated code.			
Translates high level language source code.			
Executes high level code until either the program has finished, or a syntax error is encountered.			
Translates low level language source code into machine code.			

(b) Luca is testing the computer game.

The computer that he is using is an older model with a processor that runs at a low clock speed.

Explain **one** drawback of testing the game using this computer.

(2)

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7. Question 5 · part b · 2 marks

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Further practice · Attempt part b only.

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(c) Anna uses utility software.

Complete the table by adding the type of utility software she should use to solve each issue.

Issue	Utility software
Her word processor cannot read all of a file because it is corrupt.	
She has a JPEG image and needs to send it to a publisher who can only read PNG image files.	
There has been a flood and files on a server have been damaged, but Anna needs to access her data.	
Her hard drive is running very slowly when she tries to access a file.	

(4)

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7. Question 5 · part b · 2 marks · Working and answer

Working and answer

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8. Question 2 · part c.i · 3 marks

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Further practice · Attempt part c.i only.

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2 Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

(2)

1 .....

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2 .....

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(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

(1)

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

(2)

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8. Question 2 · part c.i · 3 marks

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Further practice · Attempt part c.i only.

(c) The CPU performs the fetch-decode-execute cycle.

(i) Give **three** factors that affect the performance of the CPU.

(3)

1 .....

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2 .....

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3 .....

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(ii) Describe how the address bus is used during the fetch-decode-execute cycle.

(2)

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(d) Classical computers store data using bits that exist in one of two binary states, 1 or 0.

Quantum computers store data in a superposition of states.

Give **two other** characteristics of quantum computers.

(2)

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2 .....

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(Total for Question 2 = 12 marks)

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8. Question 2 · part c.i · 3 marks · Working and answer

Working and answer

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